Federated LLM Fine-Tuning with Split Tucker Aggregation: Framework and Beam Prediction (Supplementary Material)

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COMPLETE PROOF OF THEOREM 1

This document expands the proof sketch in Appendix A of the manuscript. We first adopt assumptions on objective smoothness, stochastic gradients, and inter-BS gradient heterogeneity, as commonly considered in federated learning analyses [1]–[3]. We then prove its bound for $\mathcal{M}\left(\bar{\Theta}^{(q,e)}\right)$ based on Assumptions 1-3, and then derive its bound for $\widehat{\mathcal{M}}\left(\bar{\Theta}^{(q,e)}\right)$ if Assumptions 4 and 5 also hold.

A. Step 1: Weighted Block Geometry and Global Smoothness

For each parameter block, $\langle\cdot,\cdot\rangle$ denotes the product-space inner product[1] obtained by summing the Euclidean or Frobenius inner products of its constituent parameters. The corresponding norm is denoted by $\|\cdot\|$.

Let $\mathbf{A} \triangleq (\mathbf{A}_{\text{fed}}, \{\mathbf{A}_{k,\text{loc}}\}_{k=1}^{K})$ and $\mathbf{B} \triangleq (\mathbf{B}_{\text{fed}}, \{\mathbf{B}_{k,\text{loc}}\}_{k=1}^{K})$ be two block tuples with the same structure as Θ. To match the averaging structure of the global objective and the BS-specific local updates, we define the weighted block inner product as

$$\langle \mathbf{A}, \mathbf{B}\rangle_{\text{blk}} \triangleq \langle \mathbf{A}_{\text{fed}}, \mathbf{B}_{\text{fed}}\rangle + \frac{1}{K}\sum_{k=1}^{K}\langle \mathbf{A}_{k,\text{loc}}, \mathbf{B}_{k,\text{loc}}\rangle. \tag{S.1}$$

Therefore, the corresponding induced norm is

$$\|\mathbf{A}\|_{\text{blk}}^2 = \langle \mathbf{A}, \mathbf{A}\rangle_{\text{blk}} = \|\mathbf{A}_{\text{fed}}\|^2 + \frac{1}{K}\sum_{k=1}^{K}\|\mathbf{A}_{k,\text{loc}}\|^2. \tag{S.2}$$

For a perturbation $\Delta\Theta = \tilde{\Theta} - \Theta$ with $\tilde{\Theta} = (\tilde{\Theta}_{\text{fed}}, \{\tilde{\Theta}_{k,\text{loc}}\}_{k=1}^{K})$, i.e., $\Delta\Theta = (\Delta\Theta_{\text{fed}}, \{\Delta\Theta_{k,\text{loc}}\}_{k=1}^{K})$ with corresponding blocks of $\Delta\Theta_{\text{fed}}$ and $\Delta\Theta_{k,\text{loc}}$. Since $F(\Theta)$ is the average of the local objectives $F_k(\Theta_k)$, its directional derivative in the direction $\Delta\Theta$ is

$$\begin{aligned}\mathrm{D}F(\Theta)[\Delta\Theta] &= \frac{1}{K}\sum_{k=1}^{K}\left(\langle \nabla_{\Theta_{\text{fed}}} F_k(\Theta_k), \Delta\Theta_{\text{fed}}\rangle + \langle \nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k), \Delta\Theta_{k,\text{loc}}\rangle\right)\\ &= \langle \nabla_{\Theta_{\text{fed}}} F(\Theta), \Delta\Theta_{\text{fed}}\rangle + \frac{1}{K}\sum_{k=1}^{K}\langle \nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k), \Delta\Theta_{k,\text{loc}}\rangle,\end{aligned} \tag{S.3}$$

where $\mathrm{D}F(\Theta)[\Delta\Theta]$ denotes the differential of F at Θ applied to $\Delta\Theta$. The second equality follows from linearity of differentiation and the fact that the shared perturbation $\Delta\Theta_{\text{fed}}$ is the same among all K BS clients. Specifically,

$$\nabla_{\Theta_{\text{fed}}} F(\Theta) = \frac{1}{K}\sum_{k=1}^{K}\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k). \tag{S.4}$$

With the inner product defined in (S.1), (S.3) can be rewritten as

$$\mathrm{D}F(\Theta)[\Delta\Theta] = \left\langle \left(\nabla_{\Theta_{\text{fed}}} F(\Theta), \{\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k)\}_{k=1}^{K}\right), \Delta\Theta\right\rangle_{\text{blk}}. \tag{S.5}$$

[1]This notation is used only in this supplement.

Therefore, the gradient with respect to this inner product can be represented by

$$\nabla_{\text{blk}} F(\Theta) = \left(\nabla_{\Theta_{\text{fed}}} F(\Theta), \{\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k)\}_{k=1}^K\right). \tag{S.6}$$

Using the definitions in (S.2) and (S.6), we obtain

$$\|\nabla_{\text{blk}} F(\Theta)\|_{\text{blk}}^2 = \|\nabla_{\Theta_{\text{fed}}} F(\Theta)\|^2 + \frac{1}{K}\sum_{k=1}^K \|\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k)\|^2 = \mathcal{M}(\Theta). \tag{S.7}$$

Let $\Delta\Theta_k \triangleq (\Delta\Theta_{\text{fed}}, \Delta\Theta_{k,\text{loc}})$. Applying the fundamental theorem of calculus to $F_k(\Theta_k + t\Delta\Theta_k)$ over $t \in [0,1]$, we obtain

$$F_k(\Theta_k + \Delta\Theta_k) - F_k(\Theta_k) = \int_0^1 \langle \nabla F_k(\Theta_k + t\Delta\Theta_k), \Delta\Theta_k\rangle \, \mathrm{d}t. \tag{S.8}$$

The first-order approximation to the change $F_k(\Theta_k + \Delta\Theta_k) - F_k(\Theta_k)$ is $\langle \nabla F_k(\Theta_k), \Delta\Theta_k\rangle$. Subtracting it from (S.8), and applying the Cauchy-Schwarz inequality and Assumption 1 yields

$$\begin{aligned}
&F_k(\Theta_k + \Delta\Theta_k) - F_k(\Theta_k) - \langle \nabla F_k(\Theta_k), \Delta\Theta_k\rangle \\
&= \int_0^1 \langle \nabla F_k(\Theta_k + t\Delta\Theta_k) - \nabla F_k(\Theta_k), \Delta\Theta_k\rangle \, \mathrm{d}t \\
&\le \int_0^1 \|\nabla F_k(\Theta_k + t\Delta\Theta_k) - \nabla F_k(\Theta_k)\| \|\Delta\Theta_k\| \, \mathrm{d}t \\
&\le \int_0^1 L_F \|t\Delta\Theta_k\| \|\Delta\Theta_k\| \, \mathrm{d}t \\
&= L_F \|\Delta\Theta_k\|^2 \int_0^1 t \, \mathrm{d}t = \frac{L_F}{2}\|\Delta\Theta_k\|^2.
\end{aligned} \tag{S.9}$$

Since the perturbation of the shared block is identical in every BS client pair, such that

$$\frac{1}{K}\sum_k \|\Delta\Theta_k\|^2 = \|\Delta\Theta_{\text{fed}}\|^2 + \frac{1}{K}\sum_k \|\Delta\Theta_{k,\text{loc}}\|^2 = \|\Delta\Theta\|_{\text{blk}}^2, \tag{S.10}$$

Averaging the local inequalities explicitly gives

$$\begin{aligned}
F(\Theta + \Delta\Theta) - F(\Theta) &= \frac{1}{K}\sum_{k=1}^K \left(F_k(\Theta_k + \Delta\Theta_k) - F_k(\Theta_k)\right) \\
&\le \frac{1}{K}\sum_{k=1}^K \langle \nabla F_k(\Theta_k), \Delta\Theta_k\rangle + \frac{L_F}{2K}\sum_{k=1}^K \|\Delta\Theta_k\|^2 \\
&= \left\langle \frac{1}{K}\sum_{k=1}^K \nabla_{\Theta_{\text{fed}}} F_k(\Theta_k), \Delta\Theta_{\text{fed}} \right\rangle + \frac{1}{K}\sum_{k=1}^K \left\langle \nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k), \Delta\Theta_{k,\text{loc}}\right\rangle + \frac{L_F}{2}\|\Delta\Theta\|_{\text{blk}}^2 \\
&= \langle \nabla_{\text{blk}} F(\Theta), \Delta\Theta\rangle_{\text{blk}} + \frac{L_F}{2}\|\Delta\Theta\|_{\text{blk}}^2.
\end{aligned} \tag{S.11}$$

Equivalently,

$$F(\Theta + \Delta\Theta) \le F(\Theta) + \langle \nabla_{\text{blk}} F(\Theta), \Delta\Theta\rangle_{\text{blk}} + \frac{L_F}{2}\|\Delta\Theta\|_{\text{blk}}^2. \tag{S.12}$$

B. Step 2: Auxiliary Update

At local step (q, e), we collect the stochastic update directions into a block tuple as

$$\widehat{\mathbf{d}}^{(q,e)} \triangleq \left(\frac{1}{K}\sum_{k=1}^K \widehat{\nabla}_{k,\text{fed}}^{(q,e)}, \{\widehat{\nabla}_{k,\text{loc}}^{(q,e)}\}_{k=1}^K\right). \tag{S.13}$$

The simultaneous stochastic gradient descent (SGD) updates at BS k are

$$\Theta_{k,\text{fed}}^{(q,e+1)} = \Theta_{k,\text{fed}}^{(q,e)} - \eta \widehat{\nabla}_{k,\text{fed}}^{(q,e)}, \tag{S.14}$$

$$\Theta_{k,\text{loc}}^{(q,e+1)} = \Theta_{k,\text{loc}}^{(q,e)} - \eta \widehat{\nabla}_{k,\text{loc}}^{(q,e)}. \tag{S.15}$$

Both gradients are evaluated at $\Theta_k^{(q,e)}$. Averaging only the first update of (S.14) yields

$$\bar{\Theta}_{\text{fed}}^{(q,e+1)} = \frac{1}{K}\sum_{k=1}^{K} \Theta_{k,\text{fed}}^{(q,e+1)} = \frac{1}{K}\sum_{k=1}^{K} \left(\Theta_{k,\text{fed}}^{(q,e)} - \eta \widehat{\nabla}_{k,\text{fed}}^{(q,e)}\right) = \bar{\Theta}_{\text{fed}}^{(q,e)} - \eta \frac{1}{K}\sum_{k=1}^{K} \widehat{\nabla}_{k,\text{fed}}^{(q,e)}. \tag{S.16}$$

Therefore, equations (S.15) and (S.16) yield the following update identity

$$\bar{\Theta}^{(q,e+1)} = \bar{\Theta}^{(q,e)} - \eta \widehat{\mathbf{d}}^{(q,e)}. \tag{S.17}$$

The auxiliary iterate is the one defined in Section V (Convergence analysis); only the federated copies are averaged, while every BS-local block follows its own update.

C. Step 3: Conditional Gradient Mismatch

Define the disagreement of the shared copies by

$$\delta_{\text{fed}}^{(q,e)} = \frac{1}{K}\sum_{k=1}^{K} \|\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)}\|^2. \tag{S.18}$$

Let $\mathbb{E}_{q,e}[\cdot] = \mathbb{E}[\cdot \mid \mathcal{H}_{q,e}]$ denote conditional expectation with respect to the history used in Assumption 2. Conditional on $\mathcal{H}_{q,e}$, the parameter iterates are fixed, while the mini-batches used at step (q,e) and the resulting stochastic gradients remain random. For each client, introduce the reference parameter tuple

$$\widetilde{\Theta}_k^{(q,e)} = \left(\bar{\Theta}_{\text{fed}}^{(q,e)}, \Theta_{k,\text{loc}}^{(q,e)}\right). \tag{S.19}$$

Since $\Theta_k^{(q,e)}$ and $\widetilde{\Theta}_k^{(q,e)}$ have the same BS-local parameter block, we have

$$\Theta_k^{(q,e)} - \widetilde{\Theta}_k^{(q,e)} = \left(\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)}, 0\right). \tag{S.20}$$

Applying the L_F-smoothness condition in Assumption 1 to these two parameter tuples yields

$$\left\|\nabla F_k(\Theta_k^{(q,e)}) - \nabla F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 \le L_F^2 \left\|\Theta_k^{(q,e)} - \widetilde{\Theta}_k^{(q,e)}\right\|^2 = L_F^2 \left\|\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)}\right\|^2. \tag{S.21}$$

By conditional unbiasedness in Assumption 2, we have

$$\mathbb{E}_{q,e}\left[\widehat{\mathbf{d}}^{(q,e)}\right] = \left(\frac{1}{K}\sum_{k=1}^{K} \nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,e)}), \left\{\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k^{(q,e)})\right\}_{k=1}^{K}\right). \tag{S.22}$$

Since BS k has parameter tuple $\widetilde{\Theta}_k^{(q,e)}$ at the auxiliary iterate $\bar{\Theta}^{(q,e)}$, evaluating (S.6) at this iterate gives

$$\nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)}) = \left(\frac{1}{K}\sum_{k=1}^{K} \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,e)}), \left\{\nabla_{\Theta_{k,\text{loc}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\}_{k=1}^{K}\right). \tag{S.23}$$

Thus, define the conditional gradient mismatch as

$$\mathbf{b}^{(q,e)} \triangleq \mathbb{E}_{q,e}\left[\widehat{\mathbf{d}}^{(q,e)}\right] - \nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)}). \tag{S.24}$$

Using the weighted block norm and Jensen's inequality for its federated component, we obtain

$$\begin{aligned}
\|\mathbf{b}^{(q,e)}\|_{\text{blk}}^2 &= \left\| \frac{1}{K}\sum_{k=1}^{K}\left[\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right]\right\|^2 + \frac{1}{K}\sum_{k=1}^{K}\left\|\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{k,\text{loc}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 \\
&\le \frac{1}{K}\sum_{k=1}^{K}\left\|\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 + \frac{1}{K}\sum_{k=1}^{K}\left\|\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{k,\text{loc}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 \\
&= \frac{1}{K}\sum_{k=1}^{K}\left(\left\|\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 + \left\|\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k^{(q,e)}) - \nabla_{\Theta_{k,\text{loc}}} F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2\right) \\
&= \frac{1}{K}\sum_{k=1}^{K}\left\|\nabla F_k(\Theta_k^{(q,e)}) - \nabla F_k(\widetilde{\Theta}_k^{(q,e)})\right\|^2 \\
&\le \frac{L_F^2}{K}\sum_{k=1}^{K}\left\|\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)}\right\|^2 \\
&= L_F^2\,\delta_{\text{fed}}^{(q,e)},
\end{aligned} \tag{S.25}$$

where the final inequality follows by applying the joint L_F-smoothness bound to the full gradient difference.

D. Step 4: Second Moment of the Stochastic Update Direction

At local step (q,e), the gradient computed from a mini-batch may differ from the gradient of F_k evaluated at the same parameter tuple. Define their difference as

$$\boldsymbol{\epsilon}_k^{(q,e)} \triangleq \widehat{\mathbf{d}}_k^{(q,e)} - \nabla F_k(\Theta_k^{(q,e)}). \tag{S.26}$$

Its federated and BS-local components are denoted by $\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,e)}$ and $\boldsymbol{\epsilon}_{k,\text{loc}}^{(q,e)}$, respectively. Collecting the stochastic gradient errors in the auxiliary update, we define

$$\boldsymbol{\nu}^{(q,e)} \triangleq \left(\frac{1}{K}\sum_{k=1}^{K}\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,e)}, \left\{\boldsymbol{\epsilon}_{k,\text{loc}}^{(q,e)}\right\}_{k=1}^{K}\right). \tag{S.27}$$

By the conditional unbiasedness and bounded variance conditions in Assumption 2, the stochastic gradient error satisfies

$$\mathbb{E}_{q,e}[\boldsymbol{\epsilon}_k^{(q,e)}] = 0, \qquad \mathbb{E}_{q,e}[\|\boldsymbol{\epsilon}_k^{(q,e)}\|^2] \le \sigma^2, \tag{S.28}$$

and then we have $\mathbb{E}_{q,e}[\boldsymbol{\nu}^{(q,e)}] = 0$.

By the definitions of $\boldsymbol{\nu}^{(q,e)}$ and $\mathbf{b}^{(q,e)}$,

$$\widehat{\mathbf{d}}^{(q,e)} = \mathbb{E}_{q,e}[\widehat{\mathbf{d}}^{(q,e)}] + \boldsymbol{\nu}^{(q,e)} = \nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)} + \boldsymbol{\nu}^{(q,e)}. \tag{S.29}$$

By the definition of the weighted block norm and Jensen's inequality,

$$\begin{aligned}
\mathbb{E}_{q,e}\left[\|\boldsymbol{\nu}^{(q,e)}\|_{\text{blk}}^2\right] &= \mathbb{E}_{q,e}\left[\left\|\frac{1}{K}\sum_{k=1}^{K}\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,e)}\right\|^2 + \frac{1}{K}\sum_{k=1}^{K}\|\boldsymbol{\epsilon}_{k,\text{loc}}^{(q,e)}\|^2\right] \\
&\le \frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{q,e}\left[\|\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,e)}\|^2 + \|\boldsymbol{\epsilon}_{k,\text{loc}}^{(q,e)}\|^2\right] \\
&= \frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{q,e}\left[\|\boldsymbol{\epsilon}_k^{(q,e)}\|^2\right] \le \sigma^2.
\end{aligned} \tag{S.30}$$

Both $\nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)})$ and $\mathbf{b}^{(q,e)}$ are measurable with respect to $\mathcal{H}_{q,e}$. Since $\mathbb{E}_{q,e}[\boldsymbol{\nu}^{(q,e)}] = 0$, we then have

$$\mathbb{E}_{q,e}\left[\left\langle \nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)}, \boldsymbol{\nu}^{(q,e)}\right\rangle_{\text{blk}}\right] = \left\langle \nabla_{\text{blk}} F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)}, \mathbb{E}_{q,e}[\boldsymbol{\nu}^{(q,e)}]\right\rangle_{\text{blk}} = 0. \tag{S.31}$$

Thereby, expanding the squared norm in (S.29), we obtain

$$\mathbb{E}_{q,e}\left[\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2\right] = \left\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)}\right\|_{\text{blk}}^2 + \mathbb{E}_{q,e}\left[\|\boldsymbol{\nu}^{(q,e)}\|_{\text{blk}}^2\right]. \tag{S.32}$$

Expanding the first term on the right-hand side of (S.32), and applying the Cauchy-Schwarz and Young inequalities to the cross term yields

$$\begin{aligned}\left\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)}\right\|_{\text{blk}}^2 &= \|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}^2 + 2\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \mathbf{b}^{(q,e)}\right\rangle_{\text{blk}} + \|\mathbf{b}^{(q,e)}\|_{\text{blk}}^2 \\ &\le 2\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}^2 + 2\|\mathbf{b}^{(q,e)}\|_{\text{blk}}^2 \\ &\le 2\mathcal{M}(\bar{\Theta}^{(q,e)}) + 2L_F^2\delta_{\text{fed}}^{(q,e)}.\end{aligned} \tag{S.33}$$

By combining (S.30), (S.32), and (S.33), we have

$$\mathbb{E}_{q,e}\left[\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2\right] \le a_1\mathcal{M}(\bar{\Theta}^{(q,e)}) + a_2\sigma^2 + a_3L_F^2\delta_{\text{fed}}^{(q,e)}, \tag{S.34}$$

where $a_1, a_2, a_3 > 0$ are numerical constants independent of Q, E, and η.

E. Step 5: Expected Objective Change at a Local Step

At local step (q,e), applying (S.12) with $\Delta\Theta = -\eta\widehat{\mathbf{d}}^{(q,e)}$ and using (S.17) gives

$$F(\bar{\Theta}^{(q,e+1)}) \le F(\bar{\Theta}^{(q,e)}) - \eta\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \widehat{\mathbf{d}}^{(q,e)}\right\rangle_{\text{blk}} + \frac{L_F\eta^2}{2}\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2. \tag{S.35}$$

The auxiliary iterate $\bar{\Theta}^{(q,e)}$ is determined by $\mathcal{H}_{q,e}$. Taking conditional expectation and using (S.24) gives

$$\begin{aligned}\mathbb{E}_{q,e}\left[F(\bar{\Theta}^{(q,e+1)})\right] &\le F(\bar{\Theta}^{(q,e)}) - \eta\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \mathbb{E}_{q,e}[\widehat{\mathbf{d}}^{(q,e)}]\right\rangle_{\text{blk}} + \frac{L_F\eta^2}{2}\mathbb{E}_{q,e}\left[\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2\right] \\ &= F(\bar{\Theta}^{(q,e)}) - \eta\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}) + \mathbf{b}^{(q,e)}\right\rangle_{\text{blk}} + \frac{L_F\eta^2}{2}\mathbb{E}_{q,e}\left[\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2\right] \\ &= F(\bar{\Theta}^{(q,e)}) - \eta\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}^2 - \eta\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \mathbf{b}^{(q,e)}\right\rangle_{\text{blk}} + \frac{L_F\eta^2}{2}\mathbb{E}_{q,e}\left[\|\widehat{\mathbf{d}}^{(q,e)}\|_{\text{blk}}^2\right].\end{aligned} \tag{S.36}$$

By using the Cauchy-Schwarz inequality, the cross term in (S.36) leads to

$$\begin{aligned}-\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \mathbf{b}^{(q,e)}\right\rangle_{\text{blk}} &\le \left|\left\langle\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)}), \mathbf{b}^{(q,e)}\right\rangle_{\text{blk}}\right| \\ &\le \|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}\|\mathbf{b}^{(q,e)}\|_{\text{blk}} \\ &\le \frac{1}{2}\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}^2 + \frac{1}{2}\|\mathbf{b}^{(q,e)}\|_{\text{blk}}^2.\end{aligned} \tag{S.37}$$

Substituting (S.37), (S.25), and (S.34) into (S.36), and using $\|\nabla_{\text{blk}}F(\bar{\Theta}^{(q,e)})\|_{\text{blk}}^2 = \mathcal{M}(\bar{\Theta}^{(q,e)})$, we have

$$\begin{aligned}\mathbb{E}_{q,e}\left[F(\bar{\Theta}^{(q,e+1)})\right] &\le F(\bar{\Theta}^{(q,e)}) - \eta\mathcal{M}(\bar{\Theta}^{(q,e)}) + \frac{\eta}{2}\mathcal{M}(\bar{\Theta}^{(q,e)}) + \frac{\eta L_F^2}{2}\delta_{\text{fed}}^{(q,e)} \\ &\quad + \frac{L_F\eta^2}{2}\left(a_1\mathcal{M}(\bar{\Theta}^{(q,e)}) + a_2\sigma^2 + a_3L_F^2\delta_{\text{fed}}^{(q,e)}\right) \\ &= F(\bar{\Theta}^{(q,e)}) - \eta\left(\frac{1}{2} - \frac{a_1L_F\eta}{2}\right)\mathcal{M}(\bar{\Theta}^{(q,e)}) + \frac{a_2L_F\eta^2}{2}\sigma^2 + \eta\left(\frac{1}{2} + \frac{a_3L_F\eta}{2}\right)L_F^2\delta_{\text{fed}}^{(q,e)} \\ &\quad L_F^2\delta_{\text{fed}}^{(q,e)} \\ &\le F(\bar{\Theta}^{(q,e)}) - \eta\frac{1 - a_1c}{2}\mathcal{M}(\bar{\Theta}^{(q,e)}) + \frac{a_2L_F\eta^2}{2}\sigma^2 + \eta\frac{1 + a_3c}{2}L_F^2\delta_{\text{fed}}^{(q,e)} \\ &= F(\bar{\Theta}^{(q,e)}) - a_4\eta\mathcal{M}(\bar{\Theta}^{(q,e)}) + a_5L_F\eta^2\sigma^2 + a_6\eta L_F^2\delta_{\text{fed}}^{(q,e)},\end{aligned} \tag{S.38}$$

where $c > 0$ is a constant chosen such that $a_1c < 1$, and $a_4 \triangleq \frac{1-a_1c}{2}, a_5 \triangleq \frac{a_2}{2}, a_6 \triangleq \frac{1+a_3c}{2}$ are independent of Q, E, and η.

Taking expectations on both sides yields

$$\mathbb{E}\left[F(\bar{\Theta}^{(q,e+1)})\right] \le \mathbb{E}\left[F(\bar{\Theta}^{(q,e)})\right] - a_4\eta\mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)})\right] + a_5L_F\eta^2\sigma^2 + a_6\eta L_F^2\mathbb{E}\left[\delta_{\text{fed}}^{(q,e)}\right], \tag{S.39}$$

F. Step 6: Federated Parameter Disagreement and Gradient Dispersion

It remains to bound the federated parameter disagreement among the local copies of the federated parameters. Define

$$\bar{\widehat{\nabla}}_{\text{fed}}^{(q,e)} \triangleq \frac{1}{K}\sum_{k=1}^{K}\widehat{\nabla}_{k,\text{fed}}^{(q,e)}. \tag{S.40}$$

The federated copies are synchronized at the beginning of each communication round, such that $\Theta_{k,\text{fed}}^{(q,0)} = \bar{\Theta}_{\text{fed}}^{(q,0)}$ and $\delta_{\text{fed}}^{(q,0)} = 0$. Subtracting the averaged federated update from the update at BS k, we obtain

$$\Theta_{k,\text{fed}}^{(q,e+1)} - \bar{\Theta}_{\text{fed}}^{(q,e+1)} = \Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)} - \eta\left(\widehat{\nabla}_{k,\text{fed}}^{(q,e)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,e)}\right). \tag{S.41}$$

Rearranging (S.41) and summing over $j = 0, \ldots, e-1$ gives

$$\sum_{j=1}^{e}\left(\Theta_{k,\text{fed}}^{(q,j)} - \bar{\Theta}_{\text{fed}}^{(q,j)}\right) - \sum_{j=0}^{e-1}\left(\Theta_{k,\text{fed}}^{(q,j)} - \bar{\Theta}_{\text{fed}}^{(q,j)}\right) = -\eta\sum_{j=0}^{e-1}\left(\widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right). \tag{S.42}$$

After canceling the intermediate terms on the left-hand side, we obtain

$$\left(\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)}\right) - \left(\Theta_{k,\text{fed}}^{(q,0)} - \bar{\Theta}_{\text{fed}}^{(q,0)}\right) = -\eta\sum_{j=0}^{e-1}\left(\widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right). \tag{S.43}$$

Since the federated copies are synchronized at the beginning of the round, i.e., $\Theta_{k,\text{fed}}^{(q,0)} - \bar{\Theta}_{\text{fed}}^{(q,0)} = 0$. Therefore, we have

$$\Theta_{k,\text{fed}}^{(q,e)} - \bar{\Theta}_{\text{fed}}^{(q,e)} = -\eta\sum_{j=0}^{e-1}\left(\widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right). \tag{S.44}$$

We next bound the dispersion of the stochastic federated gradients on the right-hand side of (S.44).

Fix a local step (q, j) and condition on $\mathcal{H}_{q,j}$. Let

$$\bar{\boldsymbol{\epsilon}}_{\text{fed}}^{(q,j)} \triangleq \frac{1}{K}\sum_{i=1}^{K}\boldsymbol{\epsilon}_{i,\text{fed}}^{(q,j)}, \tag{S.45}$$

where $\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,j)}$ is the federated component of the stochastic gradient error defined in Step 4.

Using the stochastic gradient error defined in Step 4, and then adding and subtracting the gradients evaluated at $\widetilde{\Theta}_i^{(q,j)}$, we obtain

$$\begin{aligned}
\widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)} &= \widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \frac{1}{K}\sum_{i=1}^{K}\widehat{\nabla}_{i,\text{fed}}^{(q,j)} \\
&= \boldsymbol{\epsilon}_{k,\text{fed}}^{(q,j)} + \nabla_{\Theta_{\text{fed}}}F_k(\Theta_k^{(q,j)}) - \left(\bar{\boldsymbol{\epsilon}}_{\text{fed}}^{(q,j)} + \frac{1}{K}\sum_{i=1}^{K}\nabla_{\Theta_{\text{fed}}}F_i(\Theta_i^{(q,j)})\right) \\
&= \boldsymbol{\epsilon}_{k,\text{fed}}^{(q,j)} - \bar{\boldsymbol{\epsilon}}_{\text{fed}}^{(q,j)} + \left[\nabla_{\Theta_{\text{fed}}}F_k(\Theta_k^{(q,j)}) - \nabla_{\Theta_{\text{fed}}}F_k(\widetilde{\Theta}_k^{(q,j)}) - \frac{1}{K}\sum_{i=1}^{K}\left(\nabla_{\Theta_{\text{fed}}}F_i(\Theta_i^{(q,j)}) - \nabla_{\Theta_{\text{fed}}}F_i(\widetilde{\Theta}_i^{(q,j)})\right)\right] \\
&\quad + \left[\nabla_{\Theta_{\text{fed}}}F_k(\widetilde{\Theta}_k^{(q,j)}) - \frac{1}{K}\sum_{i=1}^{K}\nabla_{\Theta_{\text{fed}}}F_i(\widetilde{\Theta}_i^{(q,j)})\right].
\end{aligned} \tag{S.46}$$

Let $\{\mathbf{x}_k\}_{k=1}^K$ be a collection of parameter blocks with the same structure, and define $\bar{\mathbf{x}} = \frac{1}{K}\sum_{k=1}^K \mathbf{x}_k$. Expanding the squared norm gives

$$\begin{aligned}\frac{1}{K}\sum_{k=1}^K \|\mathbf{x}_k - \bar{\mathbf{x}}\|^2 &= \frac{1}{K}\sum_{k=1}^K \left(\|\mathbf{x}_k\|^2 - 2\langle \mathbf{x}_k, \bar{\mathbf{x}}\rangle + \|\bar{\mathbf{x}}\|^2\right) \\ &= \frac{1}{K}\sum_{k=1}^K \|\mathbf{x}_k\|^2 - 2\left\langle \frac{1}{K}\sum_{k=1}^K \mathbf{x}_k, \bar{\mathbf{x}}\right\rangle + \|\bar{\mathbf{x}}\|^2 \\ &= \frac{1}{K}\sum_{k=1}^K \|\mathbf{x}_k\|^2 - \|\bar{\mathbf{x}}\|^2 \\ &\le \frac{1}{K}\sum_{k=1}^K \|\mathbf{x}_k\|^2. \end{aligned} \tag{S.47}$$

Applying (S.47) to the first term of (S.46) and using Assumption 2 yields

$$\frac{1}{K}\sum_{k=1}^K \mathbb{E}_{q,j}\left[\left\|\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,j)} - \bar{\boldsymbol{\epsilon}}_{\text{fed}}^{(q,j)}\right\|^2\right] \le \frac{1}{K}\sum_{k=1}^K \mathbb{E}_{q,j}\left[\|\boldsymbol{\epsilon}_{k,\text{fed}}^{(q,j)}\|^2\right] \le \sigma^2. \tag{S.48}$$

Similarly, applying (S.47) to the second term of (S.46) and using Assumption 1 yields

$$\begin{aligned}&\frac{1}{K}\sum_{k=1}^K \left\|\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,j)}) - \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,j)}) - \frac{1}{K}\sum_{i=1}^K \left(\nabla_{\Theta_{\text{fed}}} F_i(\Theta_i^{(q,j)}) - \nabla_{\Theta_{\text{fed}}} F_i(\widetilde{\Theta}_i^{(q,j)})\right)\right\|^2 \\ &\le \frac{1}{K}\sum_{k=1}^K \left\|\nabla_{\Theta_{\text{fed}}} F_k(\Theta_k^{(q,j)}) - \nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,j)})\right\|^2 \\ &\le \frac{L_F^2}{K}\sum_{k=1}^K \left\|\Theta_{k,\text{fed}}^{(q,j)} - \bar{\Theta}_{\text{fed}}^{(q,j)}\right\|^2 \\ &= L_F^2 \delta_{\text{fed}}^{(q,j)}. \end{aligned} \tag{S.49}$$

For the third term of (S.46), the shared parameter block in every $\widetilde{\Theta}_k^{(q,j)}$ equals $\bar{\Theta}_{\text{fed}}^{(q,j)}$, while the local block remains specific to BS k. When differentiating the global objective with respect to the federated parameter block, the BS-local parameter blocks are held fixed. Therefore,

$$\nabla_{\Theta_{\text{fed}}} F(\bar{\Theta}^{(q,j)}) = \nabla_{\Theta_{\text{fed}}}\left[\frac{1}{K}\sum_{i=1}^K F_i\left(\bar{\Theta}_{\text{fed}}^{(q,j)}, \Theta_{i,\text{loc}}^{(q,j)}\right)\right] = \frac{1}{K}\sum_{i=1}^K \nabla_{\Theta_{\text{fed}}} F_i\left(\widetilde{\Theta}_i^{(q,j)}\right). \tag{S.50}$$

With (S.50), the third term in (S.46) satisfies

$$\begin{aligned}\frac{1}{K}\sum_{k=1}^K \left\|\nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,j)}) - \frac{1}{K}\sum_{i=1}^K \nabla_{\Theta_{\text{fed}}} F_i(\widetilde{\Theta}_i^{(q,j)})\right\|^2 &= \frac{1}{K}\sum_{k=1}^K \left\|\nabla_{\Theta_{\text{fed}}} F_k(\widetilde{\Theta}_k^{(q,j)}) - \nabla_{\Theta_{\text{fed}}} F(\bar{\Theta}^{(q,j)})\right\|^2 \\ &= \Gamma_{\text{fed}}(\bar{\Theta}^{(q,j)}) \le \Delta_{\text{fed}}^2, \end{aligned} \tag{S.51}$$

where the second equality follows from the definition of Γ_{fed}, and the inequality follows from Assumption 3.

To combine the bounds on the three components in (S.46), we use the triangle inequality followed by the Cauchy-Schwarz inequality:

$$\|\mathbf{x}+\mathbf{y}+\mathbf{z}\|^2 \le (\|\mathbf{x}\| + \|\mathbf{y}\| + \|\mathbf{z}\|)^2 \le 3\left(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \|\mathbf{z}\|^2\right). \tag{S.52}$$

Applying this bound to (S.46), averaging over the BSs, and taking conditional expectation lead to

$$\frac{1}{K}\sum_{k=1}^K \mathbb{E}_{q,j}\left[\left\|\widehat{\nabla}_{k,\text{fed}}^{(q,j)} - \bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right\|^2\right] \le 3\sigma^2 + 3L_F^2\delta_{\text{fed}}^{(q,j)} + 3\Delta_{\text{fed}}^2 \le a_7\left(\sigma^2 + \Delta_{\text{fed}}^2 + L_F^2\delta_{\text{fed}}^{(q,j)}\right), \tag{S.53}$$

where $a_7 \geq 3$ is a numerical constant independent of Q, E, and η.

Finally, taking expectations on (S.53) gives

$$\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\left[\left\|\widehat{\nabla}_{k,\text{fed}}^{(q,j)}-\bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right\|^2\right]\leq a_7\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2\mathbb{E}[\delta_{\text{fed}}^{(q,j)}]\right). \tag{S.54}$$

G. Step 7: Uniform Bound on Federated Parameter Disagreement

For any $e \geq 1$ and any equally structured parameter blocks $\{\mathbf{x}_j\}_{j=0}^{e-1}$, the Cauchy-Schwarz inequality gives

$$\left\|\sum_{j=0}^{e-1}\mathbf{x}_j\right\|^2=\sum_{i=0}^{e-1}\sum_{j=0}^{e-1}\langle\mathbf{x}_i,\mathbf{x}_j\rangle\leq\sum_{i=0}^{e-1}\sum_{j=0}^{e-1}\|\mathbf{x}_i\|\|\mathbf{x}_j\|=\left(\sum_{j=0}^{e-1}\|\mathbf{x}_j\|\right)^2\leq\left(\sum_{j=0}^{e-1}1^2\right)\left(\sum_{j=0}^{e-1}\|\mathbf{x}_j\|^2\right)\leq e\sum_{j=0}^{e-1}\|\mathbf{x}_j\|^2. \tag{S.55}$$

Substituting (S.44) into (S.18) and applying (S.55) for each BS, we obtain

$$\mathbb{E}[\delta_{\text{fed}}^{(q,e)}]=\frac{\eta^2}{K}\sum_{k=1}^{K}\mathbb{E}\left[\left\|\sum_{j=0}^{e-1}\left(\widehat{\nabla}_{k,\text{fed}}^{(q,j)}-\bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right)\right\|^2\right]\leq\eta^2 e\sum_{j=0}^{e-1}\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\left[\left\|\widehat{\nabla}_{k,\text{fed}}^{(q,j)}-\bar{\widehat{\nabla}}_{\text{fed}}^{(q,j)}\right\|^2\right]. \tag{S.56}$$

Substituting (S.54) into (S.56), we obtain

$$\mathbb{E}[\delta_{\text{fed}}^{(q,e)}]\leq a_7\eta^2 e\sum_{j=0}^{e-1}\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2\mathbb{E}[\delta_{\text{fed}}^{(q,j)}]\right). \tag{S.57}$$

For a fixed round q, define

$$d_q\triangleq\max_{0\leq e\leq E}\mathbb{E}[\delta_{\text{fed}}^{(q,e)}]. \tag{S.58}$$

This implies that for $1 \leq e \leq E$, we have $\mathbb{E}[\delta_{\text{fed}}^{(q,j)}] \leq d_q$ for every $j < e$. Therefore, (S.57) becomes

$$\mathbb{E}[\delta_{\text{fed}}^{(q,e)}]\leq a_7\eta^2 e\sum_{j=0}^{e-1}\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2 d_q\right)=a_7\eta^2e^2\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2d_q\right)\leq a_7\eta^2E^2\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2d_q\right). \tag{S.59}$$

The right-hand side is independent of e and also bounds the zero disagreement at $e = 0$. Furthermore, by the definition of d_q, we have

$$d_q\leq a_7\eta^2E^2\left(\sigma^2+\Delta_{\text{fed}}^2+L_F^2d_q\right). \tag{S.60}$$

The step-size condition $\eta \leq c/(L_F E)$ leads to $a_7 L_F^2\eta^2E^2 \leq a_7c^2$. Choose the fixed constant $c > 0$ sufficiently small that $a_1c < 1$ and $a_7c^2 < 1$. Rewriting (S.60), we have

$$\left(1-a_7L_F^2\eta^2E^2\right)d_q\leq a_7\eta^2E^2\left(\sigma^2+\Delta_{\text{fed}}^2\right). \tag{S.61}$$

Since $1 - a_7L_F^2\eta^2E^2 \geq 1 - a_7c^2 > 0$, we can divide both sides of (S.61) by $1 - a_7L_F^2\eta^2E^2$ without changing the direction of the inequality. This gives

$$d_q\leq\frac{a_7\eta^2E^2}{1-a_7L_F^2\eta^2E^2}\left(\sigma^2+\Delta_{\text{fed}}^2\right)\leq\frac{a_7}{1-a_7c^2}\eta^2E^2\left(\sigma^2+\Delta_{\text{fed}}^2\right). \tag{S.62}$$

Consequently, for every $0 \leq q < Q$ and $0 \leq e \leq E$,

$$\mathbb{E}[\delta_{\text{fed}}^{(q,e)}]\leq a_8\eta^2E^2\left(\sigma^2+\Delta_{\text{fed}}^2\right),\quad a_8\triangleq\frac{a_7}{1-a_7c^2}. \tag{S.63}$$

H. Step 8: Trainable Parameter Stationarity Bound for the Statement of (41) in Theorem 1

Substituting (S.63) into (S.39) and moving the term $a_4\eta\mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)})\right]$ to the left-hand side gives

$$a_4\eta\mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)})\right] \leq \mathbb{E}\left[F(\bar{\Theta}^{(q,e)}) - F(\bar{\Theta}^{(q,e+1)})\right] + a_5 L_F \eta^2\sigma^2 + a_6 a_8 L_F^2\eta^3 E^2\left(\sigma^2 + \Delta_{\text{fed}}^2\right). \tag{S.64}$$

At the end of round q, the server averages the federated parameter copies, while each BS carries its local parameters into the next round. Thus,

$$\bar{\Theta}_{\text{fed}}^{(q+1,0)} = \bar{\Theta}_{\text{fed}}^{(q,E)}, \quad \Theta_{k,\text{loc}}^{(q+1,0)} = \Theta_{k,\text{loc}}^{(q,E)}, \quad k = 1,\ldots,K. \tag{S.65}$$

Let $\bar{\Theta}^{(0,0)} = \Theta^{(0)}$. Summing the objective differences within each round and using (S.65), we then obtain

$$\begin{aligned}
\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1}\mathbb{E}\Big[F(\bar{\Theta}^{(q,e)}) - F(\bar{\Theta}^{(q,e+1)})\Big] &= \sum_{q=0}^{Q-1}\mathbb{E}\Big[F(\bar{\Theta}^{(q,0)}) - F(\bar{\Theta}^{(q,E)})\Big] \\
&= \sum_{q=0}^{Q-1}\mathbb{E}\left[F(\bar{\Theta}^{(q,0)}) - F(\bar{\Theta}^{(q+1,0)})\right] \\
&= F(\Theta^{(0)}) - \mathbb{E}\left[F(\bar{\Theta}^{(Q,0)})\right] \\
&\leq F(\Theta^{(0)}) - F^\star,
\end{aligned} \tag{S.66}$$

where the final inequality uses the lower bound on F stated in Assumption 1.

Summing (S.64) over $q = 0,\ldots,Q-1$ and $e = 0,\ldots,E-1$, and then applying (S.66), leads to

$$a_4\eta\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1}\mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)})\right] \leq F(\Theta^{(0)}) - F^\star + QE\,a_5 L_F\eta^2\sigma^2 + QE\,a_6 a_8 L_F^2\eta^3E^2\left(\sigma^2 + \Delta_{\text{fed}}^2\right). \tag{S.67}$$

Dividing both sides of (S.67) by $a_4\eta QE > 0$ gives

$$\frac{1}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1}\mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)})\right] \leq \frac{F(\Theta^{(0)}) - F^\star}{a_4\eta QE} + \frac{a_5}{a_4}L_F\eta\sigma^2 + \frac{a_6a_8}{a_4}L_F^2\eta^2E^2\left(\sigma^2 + \Delta_{\text{fed}}^2\right). \tag{S.68}$$

Then we can obtain (41) in the *Theorem 1*

$$\frac{1}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1}\mathbb{E}\Big[\mathcal{M}\Big(\bar{\Theta}^{(q,e)}\Big)\Big] \leq \frac{C_0(F(\Theta^{(0)}) - F^\star)}{\eta QE} + C_\sigma L_F\eta\sigma^2 + C_{\text{dr}}L_F^2\eta^2E^2\left(\sigma^2 + \Delta_{\text{fed}}^2\right), \tag{S.69}$$

where

$$C_0 \triangleq \frac{1}{a_4}, \quad C_\sigma \triangleq \frac{a_5}{a_4}, \quad C_{\text{dr}} \triangleq \frac{a_6a_8}{a_4}, \tag{S.70}$$

which are independent of Q, E, and η. This gives convergence bound stated in (41) of *Theorem 1*.

I. Step 9: Relation Between Trainable and Effective Adapter Gradients

The bound established in Step 8 concerns gradients with respect to the trainable Tucker core and factor matrices. To relate it to gradients with respect to the effective adapter updates, we use Assumptions 4 and 5. Assumption 4 bounds the component of the effective adapter gradient orthogonal to the update subspace induced jointly by the shared Tucker core and BS-local factors. Related bounds on gradient components outside low-rank tangent spaces have been used in analyses of Tucker training and low-rank fine-tuning adaptation [4], [5]. Furthermore, motivated by related comparisons between effective weight gradients and trainable factor gradients in federated low-rank adaptation (LoRA) and low-rank optimization [6], [7], Assumption 5 bounds the projected effective adapter gradient in terms of the trainable Tucker parameter gradients.

We now prove (42) based on Assumptions 4 and 5. For each adapted layer ℓ, $\mathbf{P}_{\mathbf{r},\ell}(\Theta)$ is an orthogonal projector. Thus, $\mathbf{P}_{\mathbf{r},\ell}(\Theta)^\top = \mathbf{P}_{\mathbf{r},\ell}(\Theta)$ and $\mathbf{P}_{\mathbf{r},\ell}(\Theta)^2 = \mathbf{P}_{\mathbf{r},\ell}(\Theta)$. It follows that

$$\langle\mathbf{P}_{\mathbf{r},\ell}(\Theta)\mathbf{z}_\ell(\Theta), (\mathbf{I} - \mathbf{P}_{\mathbf{r},\ell}(\Theta))\,\mathbf{z}_\ell(\Theta)\rangle = \mathbf{z}_\ell(\Theta)^\top\left(\mathbf{P}_{\mathbf{r},\ell}(\Theta) - \mathbf{P}_{\mathbf{r},\ell}(\Theta)^2\right)\mathbf{z}_\ell(\Theta) = 0. \tag{S.71}$$

The projected and residual components of $\mathbf{z}_\ell(\Theta)$ are therefore orthogonal. Using Assumptions 4 and 5, we obtain

$$\|\mathbf{z}_\ell(\Theta)\|^2 = \|\mathbf{P}_{\mathbf{r},\ell}(\Theta)\mathbf{z}_\ell(\Theta)\|^2 + \|\big(\mathbf{I} - \mathbf{P}_{\mathbf{r},\ell}(\Theta)\big)\mathbf{z}_\ell(\Theta)\|^2 \le \kappa_\ell \omega_\ell(\Theta) + \varepsilon_\ell(\mathbf{r}). \tag{S.72}$$

The Tucker cores and BS-local factor matrices associated with different adapted layers occupy disjoint parameter blocks. Under the FedTucker-S parameter partition scheme presented in Section IV, the squared gradient norms can be decomposed as

$$\|\nabla_{\Theta_{\text{fed}}} F(\Theta)\|^2 = \sum_{\ell \in \mathcal{L}_{\text{adp}}} \|\nabla_{\boldsymbol{\mathcal{G}}_\ell} F(\Theta)\|_F^2 + \|\nabla_{\Theta_{\text{out}}} F(\Theta)\|^2, \tag{S.73}$$

$$\|\nabla_{\Theta_{k,\text{loc}}} F_k(\Theta_k)\|^2 = \sum_{\ell \in \mathcal{L}_{\text{adp}}} \|\nabla_{\Theta_{k,\ell}} F_k(\Theta_k)\|^2 + \|\nabla_{\Theta_{k,\text{GPS}}} F_k(\Theta_k)\|^2 + \|\nabla_{\Theta_{k,\text{int}}} F_k(\Theta_k)\|^2 + |\nabla_{\zeta_k} F_k(\Theta_k)|^2 . \tag{S.74}$$

By the definition of $\omega_\ell(\Theta)$, summing its values over the adapted layers and using (S.73) and (S.74) gives

$$\mathcal{M}(\Theta) - \sum_{\ell \in \mathcal{L}_{\text{adp}}} \omega_\ell(\Theta) = \|\nabla_{\Theta_{\text{out}}} F(\Theta)\|^2 + \frac{1}{K}\sum_{k=1}^{K} \left(\|\nabla_{\Theta_{k,\text{GPS}}} F_k(\Theta_k)\|^2 + \|\nabla_{\Theta_{k,\text{int}}} F_k(\Theta_k)\|^2 + |\nabla_{\zeta_k} F_k(\Theta_k)|^2 \right) \ge 0. \tag{S.75}$$

Thus, $\sum_{\ell \in \mathcal{L}_{\text{adp}}} \omega_\ell(\Theta) \le \mathcal{M}(\Theta)$. Moreover, since $\omega_\ell(\Theta)$ for every $\ell \in \mathcal{L}_{\text{adp}}$ is nonnegative, we have

$$\sum_{\ell \in \mathcal{L}_{\text{adp}}} \kappa_\ell \omega_\ell(\Theta) \le \left(\max_{\ell \in \mathcal{L}_{\text{adp}}} \kappa_\ell \right) \sum_{\ell \in \mathcal{L}_{\text{adp}}} \omega_\ell(\Theta) \le \left(\max_{\ell \in \mathcal{L}_{\text{adp}}} \kappa_\ell \right) \mathcal{M}(\Theta). \tag{S.76}$$

Finally, summing (S.72) over the adapted layers and applying (S.76), we obtain

$$\begin{aligned} \widehat{\mathcal{M}}(\Theta) &= \mathcal{M}(\Theta) + \sum_{\ell \in \mathcal{L}_{\text{adp}}} \|\mathbf{z}_\ell(\Theta)\|^2 \\ &\le \mathcal{M}(\Theta) + \sum_{\ell \in \mathcal{L}_{\text{adp}}} \kappa_\ell \omega_\ell(\Theta) + \sum_{\ell \in \mathcal{L}_{\text{adp}}} \varepsilon_\ell(\mathbf{r}) \\ &\le \left(1 + \max_{\ell \in \mathcal{L}_{\text{adp}}} \kappa_\ell \right) \mathcal{M}(\Theta) + \varepsilon_{\text{T}}(\mathbf{r}) \\ &= \kappa \mathcal{M}(\Theta) + \varepsilon_{\text{T}}(\mathbf{r}), \end{aligned} \tag{S.77}$$

for (40) in Section V.

J. Step 10: Effective Adapter Gradient Bound

For the fixed rank configuration $\mathbf{r}$, suppose Assumptions 4 and 5 hold at every auxiliary iterate. Hence, applying (S.77) to every auxiliary iterate $\bar{\Theta}^{(q,e)}$ with the same constants κ and $\varepsilon_{\text{T}}(\mathbf{r})$ and taking expectations leads to

$$\mathbb{E}\left[\widehat{\mathcal{M}}(\bar{\Theta}^{(q,e)}) \right] \le \mathbb{E}\left[\kappa \mathcal{M}(\bar{\Theta}^{(q,e)}) + \varepsilon_{\text{T}}(\mathbf{r}) \right] = \kappa \mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)}) \right] + \varepsilon_{\text{T}}(\mathbf{r}). \tag{S.78}$$

Averaging (S.78) over all QE local steps yields

$$\begin{aligned} \frac{1}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1} \mathbb{E}\left[\widehat{\mathcal{M}}(\bar{\Theta}^{(q,e)}) \right] &\le \frac{\kappa}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1} \mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)}) \right] + \frac{1}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1} \varepsilon_{\text{T}}(\mathbf{r}) \\ &= \frac{\kappa}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1} \mathbb{E}\left[\mathcal{M}(\bar{\Theta}^{(q,e)}) \right] + \varepsilon_{\text{T}}(\mathbf{r}). \end{aligned} \tag{S.79}$$

Applying (S.68) to (S.79) gives

$$\begin{aligned} \frac{1}{QE}\sum_{q=0}^{Q-1}\sum_{e=0}^{E-1} \mathbb{E}\left[\widehat{\mathcal{M}}(\bar{\Theta}^{(q,e)}) \right] &\le \kappa \left[\frac{F(\Theta^{(0)}) - F^\star}{a_4 \eta Q E} + \frac{a_5}{a_4} L_F \eta \sigma^2 + \frac{a_6 a_8}{a_4} L_F^2 \eta^2 E^2 \left(\sigma^2 + \Delta_{\text{fed}}^2 \right) \right] + \varepsilon_{\text{T}}(\mathbf{r}) \\ &= \frac{\kappa C_0 \left(F(\Theta^{(0)}) - F^\star \right)}{\eta Q E} + \kappa C_\sigma L_F \eta \sigma^2 + \kappa C_{\text{dr}} L_F^2 \eta^2 E^2 \left(\sigma^2 + \Delta_{\text{fed}}^2 \right) + \varepsilon_{\text{T}}(\mathbf{r}), \end{aligned} \tag{S.80}$$

where C_0, C_σ, and C_{dr} are defined in (S.70). This gives convergence bound stated in (42) of *Theorem 1*.

This completes the proof of *Theorem 1*.

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